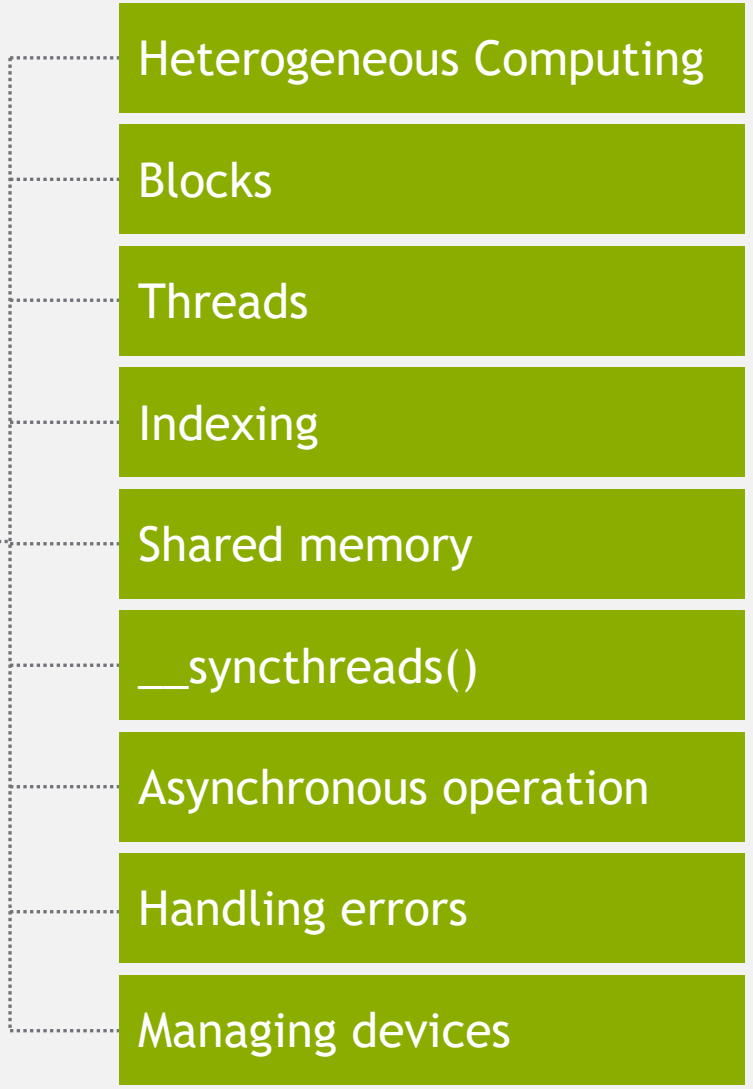
CUDA C/C++ BASICS

What is CUDA?

- **Acronym for Compute Unified Device Architecture.**
  - **A parallel computing architecture developed by NVIDIA.**
  - **The computing engine in GPU.**
  - **CUDA can be accessible to software developers through industry standard programming languages.**
- **CUDA gives developers access to the instruction set and memory of the parallel computation elements in GPUs.**

With CUDA, we can tell the GPU to perform general calculations too.

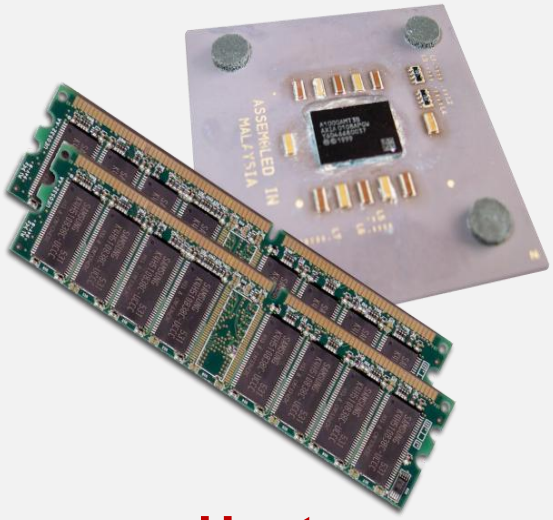
CONCEPTS


Heterogeneous Computing
Blocks
Threads
Indexing
Shared memory
__syncthreads()
Asynchronous operation
Handling errors
Managing devices

Heterogeneous Computing

Heterogeneous computing means using both CPU and GPU in the same program.

- Terminology:
  - *Host* The CPU and its memory (host memory)
  - *Device* The GPU and its memory (device memory)



Host



Device

Heterogeneous Computing

```
#include <iostream>
#include <algorithm>

using namespace std;

#define N          1024
#define RADIUS     3
#define BLOCK_SIZE 16

__global__ void stencil_1d(int *in, int *out) {
        __shared__ int temp[BLOCK_SIZE + 2 * RADIUS];
        int gindex = threadIdx.x + blockIdx.x * blockDim.x;
        int lindex = threadIdx.x + RADIUS;

        // Read input elements into shared memory
        temp[lindex] = in[gindex];
        if (threadIdx.x < RADIUS) {
                temp[lindex - RADIUS] = in[gindex - RADIUS];
                temp[lindex + BLOCK_SIZE] = in[gindex + BLOCK_SIZE];
        }

        // Synchronize (ensure all the data is available)
        __syncthreads();

        // Apply the stencil
        int result = 0;
        for (int offset = -RADIUS ; offset <= RADIUS ; offset++)
                result += temp[lindex + offset];

        // Store the result
        out[gindex] = result;
}

void fill_ints(int *x, int n) {
        fill_n(x, n, 1);
}

int main(void) {
        int *in, *out;          // host copies of a, b, c
        int *d_in, *d_out;      // device copies of a, b, c
        int size = (N + 2*RADIUS) * sizeof(int);

        // Alloc space for host copies and setup values
        in  = (int *)malloc(size); fill_ints(in,  N + 2*RADIUS);
        out = (int *)malloc(size); fill_ints(out, N + 2*RADIUS);

        // Alloc space for device copies
        cudaMalloc((void **)&d_in,  size);
        cudaMalloc((void **)&d_out, size);

        // Copy to device
        cudaMemcpy(d_in,  in,  size, cudaMemcpyHostToDevice);
        cudaMemcpy(d_out, out, size, cudaMemcpyHostToDevice);

        // Launch stencil_1d() kernel on GPU
        stencil_1d<<<N/BLOCK_SIZE,BLOCK_SIZE>>>(d_in + RADIUS,
d_out + RADIUS);

        // Copy result back to host
        cudaMemcpy(out, d_out, size, cudaMemcpyDeviceToHost);

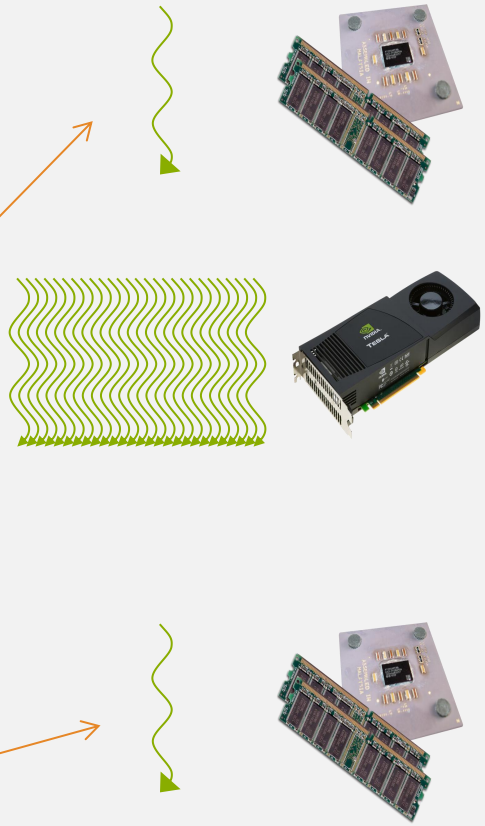
        // Cleanup
        free(in); free(out);
        cudaFree(d_in); cudaFree(d_out);
        return 0;
}
```

parallel fn

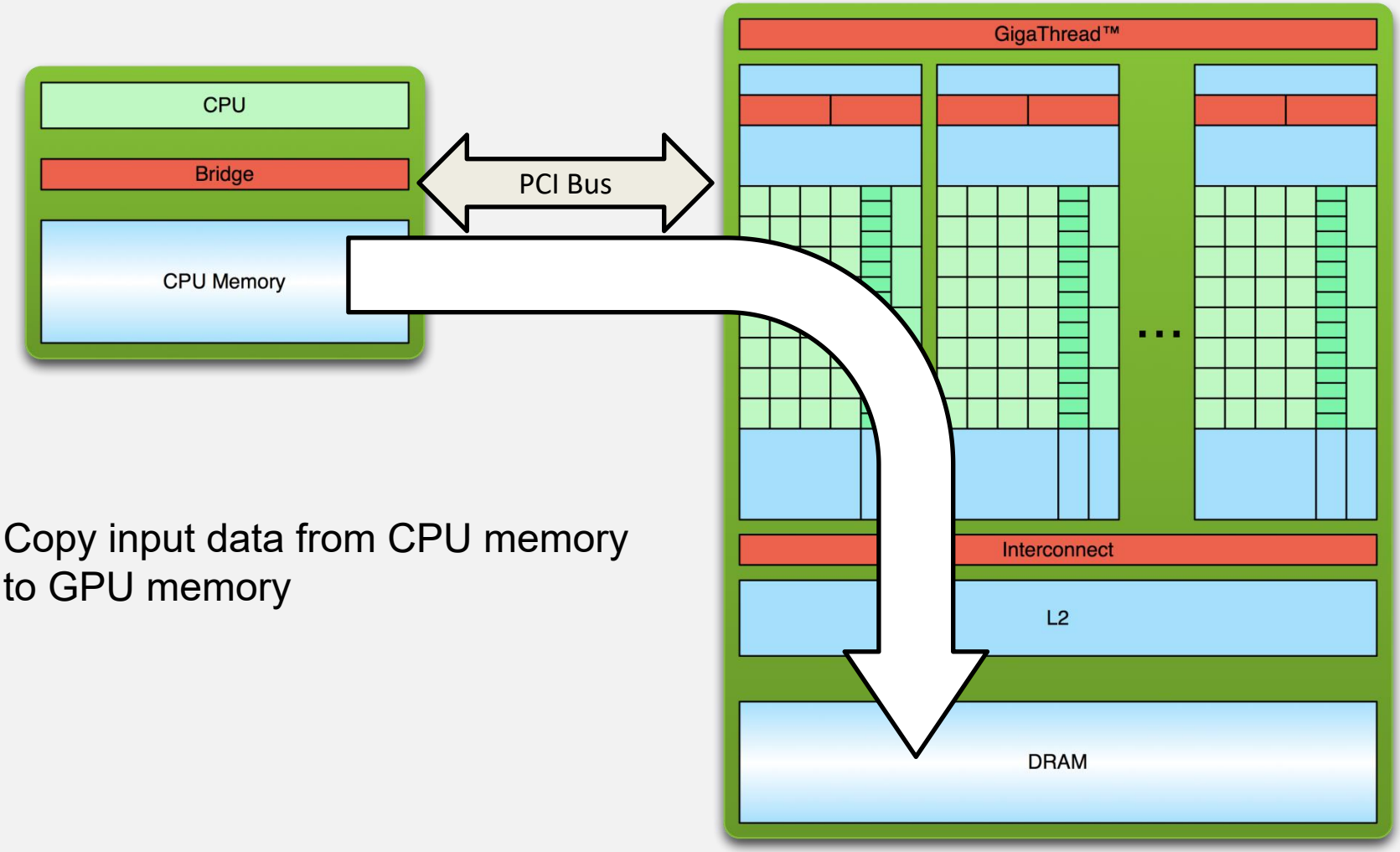
serial code

parallel code

serial code

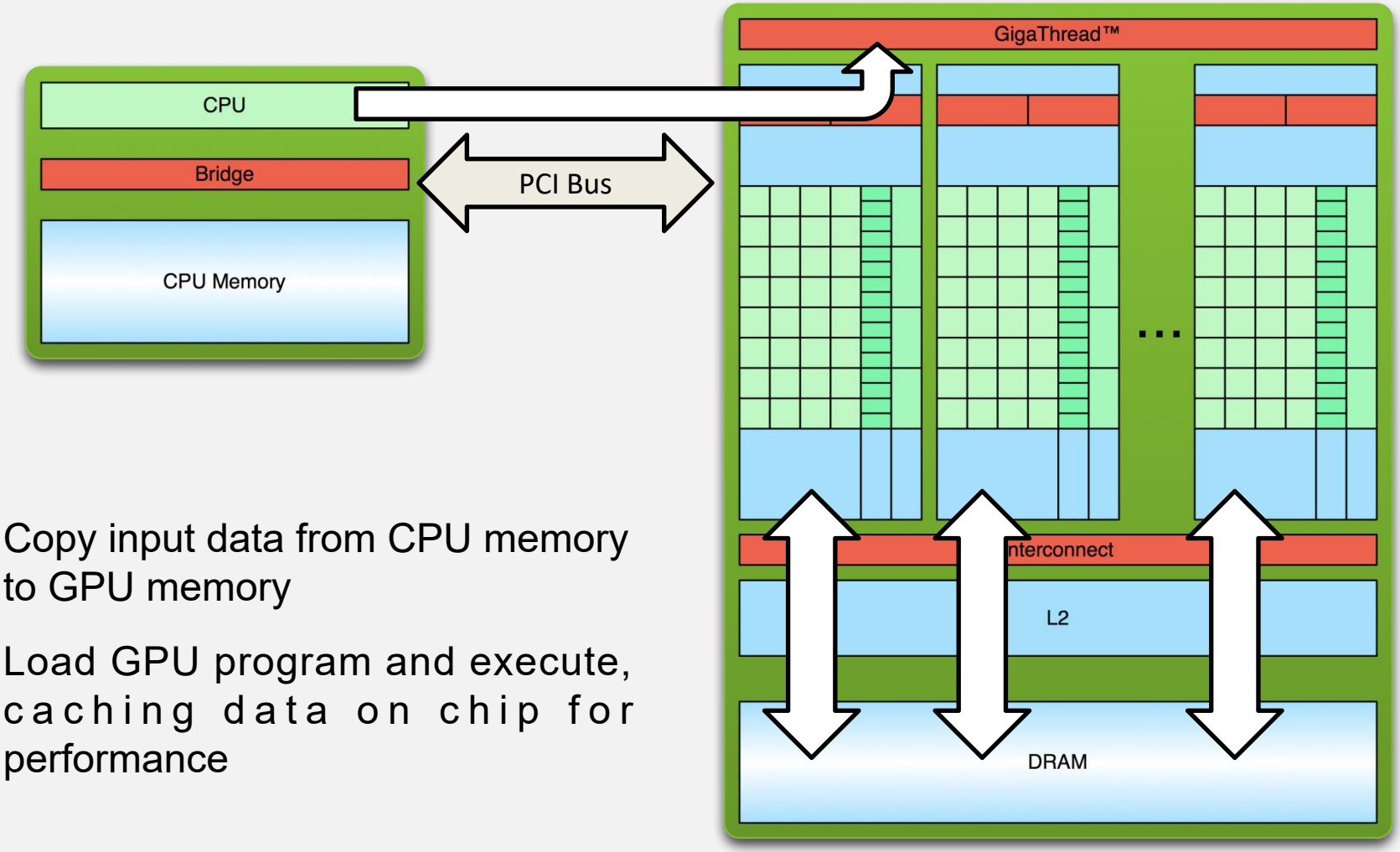


Simple Processing Flow



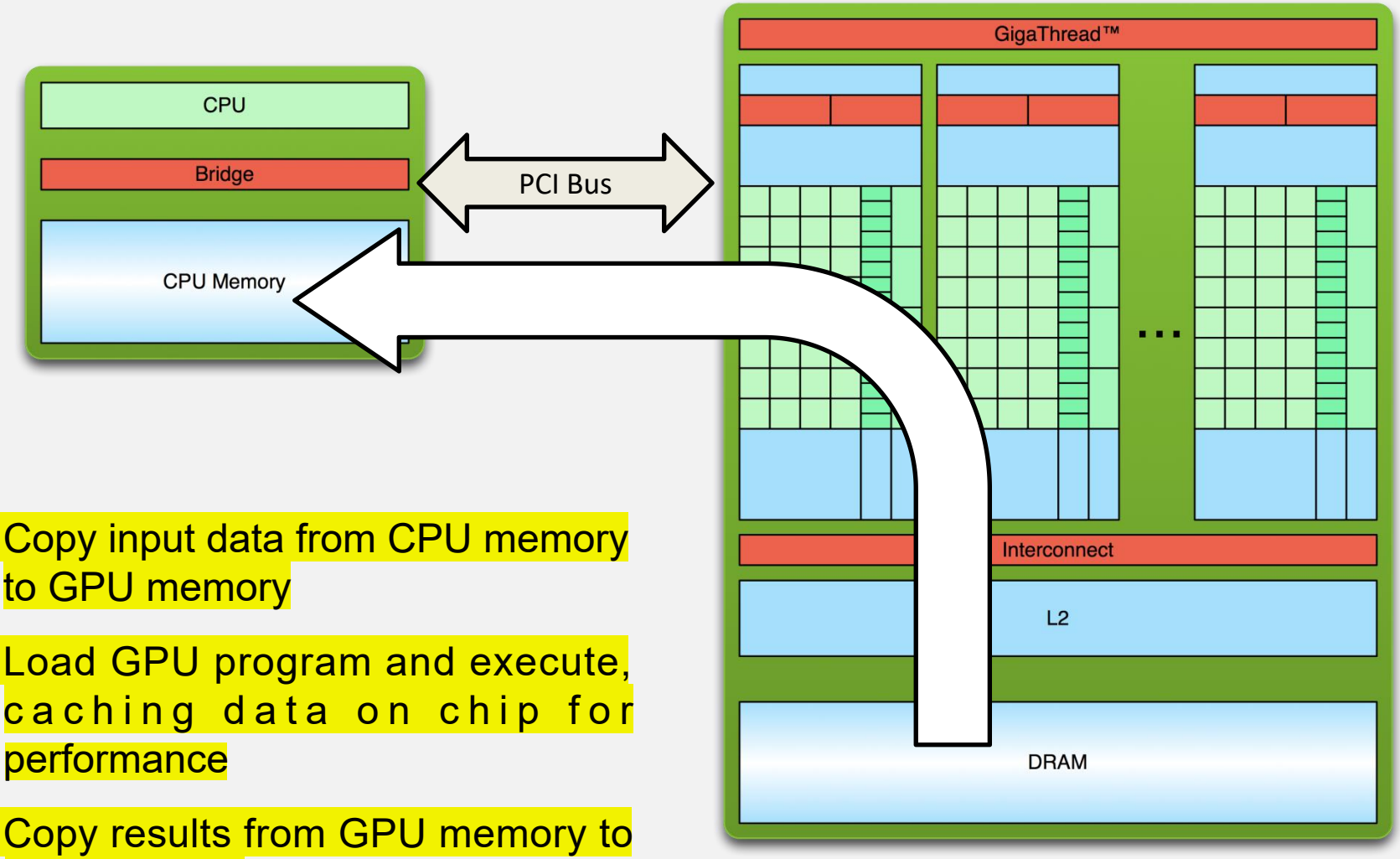
1. Copy input data from CPU memory to GPU memory

Simple Processing Flow



1. Copy input data from CPU memory to GPU memory
2. Load GPU program and execute, caching data on chip for performance

Simple Processing Flow



1. Copy input data from CPU memory to GPU memory
2. Load GPU program and execute, caching data on chip for performance
3. Copy results from GPU memory to CPU memory

© NVIDIA 2013

Hello World!

Standard C that runs on the host

```
int main(void)
{
printf("Hello World!\n");
return 0;
}
```

- **NVIDIA compiler (nvcc) can be used to compile programs with no *device* code**

Output:

```
$ nvcc
hello_world.
cu
$ a.out
Hello World!
$
```

Hello World! with Device Code

__global__ = This function will run on the GPU, but it will be called from the CPU

```
__global__ void mykernel(void) {
}

int main(void) {
        mykernel<<<1,1>>>();
        printf("Hello World!\n");
        return 0;
}
```

<< >> means = "Launch this function on the GPU." This is called a kernel launch.

kernel is a function that runs on the GPU.

- Two new syntactic elements…

Hello World! with Device Code

```
__global__ void mykernel(void) {
}
```

- CUDA C/C++ keyword `__global__` indicates a function that:
  - Runs on the device
  - Is called from host code

- `nvcc` separates source code into host and device components
  - Device functions (e.g. `mykernel()`) processed by NVIDIA compiler
  - Host functions (e.g. `main()`) processed by standard host compiler
    - `gcc`, `cl.exe`

Hello World! with Device COde

```
mykernel<<<1,1>>>();
```

- Triple angle brackets mark a call from *host* code to *device* code
  - Also called a "**kernel launch**"

Hello World! with Device Code

```
__global__ void mykernel(void){
}

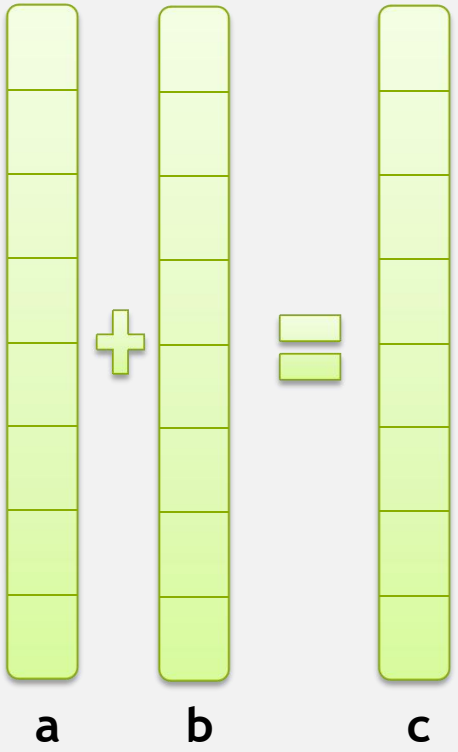
int main(void) {
        mykernel<<<1,1>>>();
        printf("Hello World!\n");
        return 0;
}
```

Output:

```
$ nvcc
hello.cu
$ a.out
Hello World!
$
```

Parallel Programming in CUDA C/C++

- But GPU computing is about massive parallelism!
- For example adding two integers and build up to vector addition



Addition on the Device

- A simple kernel to add two integers

```
__global__ void add(int *a, int *b, int *c) {
        *c = *a + *b;
}
```

- As before `__global__` is a CUDA C/C++ keyword meaning
  - `add()` will execute on the device
  - `add()` will be called from the host

Addition on the Device

- Note that we use pointers for the variables

```
__global__ void add(int *a, int *b, int *c) {
        *c = *a + *b;
}
```

- **`add()`** runs on the device, so **`a`**, **`b`** and **`c`** must point to device memory

- We need to allocate memory on the GPU

Memory Management

they can call or send receive data but cannot update other memory

- Host and device memory are separate entities
  - *Device* pointers point to GPU memory

    May be passed to/from host code

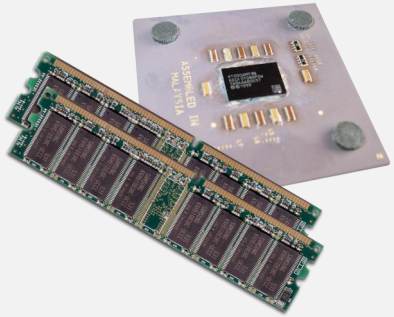
    May *not* be dereferenced in host code



  - *Host* pointers point to CPU memory

    May be passed to/from device code

    May *not* be dereferenced in device code



- Simple CUDA API for handling device memory
  - `cudaMalloc()`, `cudaFree()`, `cudaMemcpy()`
  - Similar to the C equivalents `malloc()`, `free()`, `memcpy()`

Addition on the Device: `main()`

```
__global__ void add(int *a, int *b, int *c) {
        *c = *a + *b;
}
```

```
int main(void) {
        int a, b, c;                    // host copies of a, b, c
        int *d_a, *d_b, *d_c;           // device copies of a, b, c
        int size = sizeof(int);

        // Allocate space for device copies of a, b, c
        cudaMalloc((void **)&d_a, size);
        cudaMalloc((void **)&d_b, size);
        cudaMalloc((void **)&d_c, size);

        // Setup input values
        a = 2;
        b = 7;
```

Addition on the Device: `main()`

```
    // Copy inputs to device
    cudaMemcpy(d_a, &a, size, cudaMemcpyHostToDevice);
    cudaMemcpy(d_b, &b, size, cudaMemcpyHostToDevice);

    // Launch add() kernel on GPU
    add<<<1,1>>>(d_a, d_b, d_c);

    // Copy result back to host
    cudaMemcpy(&c, d_c, size, cudaMemcpyDeviceToHost);

    // Cleanup
    cudaFree(d_a);
    cudaFree(d_b);
    cudaFree(d_c);
    return 0;
}
```

RUNNING IN PARALLEL

CONCEPTS

- Heterogeneous Computing
- Blocks
- Threads
- Indexing
- Shared memory
- __syncthreads()
- Asynchronous operation
- Handling errors
- Managing devices

Block

Moving to Parallel

- GPU computing is about massive parallelism
- Instead of executing `add()` once, execute N times in parallel

```
add<<< 1, 1 >>>();
```



```
add<<< N, 1 >>>();
```

<<<Block , Thread>>>

add<<<N,1>>>();

N blocks
1 thread per block

Vector Addition on the Device

A block is a group of GPU threads.

- With **`add()`** running in parallel we can do vector addition

- **Terminology:** each parallel invocation of **`add()`** is referred to as a block
  - The set of blocks is referred to as a grid
  - Each invocation can refer to its block index using `blockIdx.x`

The collection of blocks is called a grid.

```
__global__ void add(int *a, int *b, int *c) {
        c[blockIdx.x] = a[blockIdx.x] + b[blockIdx.x];
}
```

Every block has an ID.

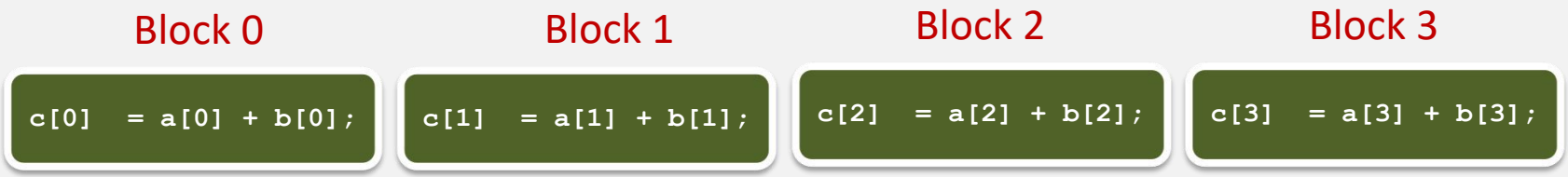
- By using `blockIdx.x` to index into the array, each block handles a different index

© NVIDIA 2013

Vector Addition on the Device: `add()`

```
__global__ void add(int *a, int *b, int *c) {
        c[blockIdx.x] = a[blockIdx.x] + b[blockIdx.x];
}
```

- On the device, each block can execute in parallel:



Vector Addition on the Device: `main()`

```
#define N 512
int main(void) {
    int *a, *b, *c;         // host copies of a, b, c
    int *d_a, *d_b, *d_c;  // device copies of a, b, c
    int size = N * sizeof(int);

    // Alloc space for device copies of a, b, c
    cudaMalloc((void **)&d_a, size);
    cudaMalloc((void **)&d_b, size);
    cudaMalloc((void **)&d_c, size);

    // Alloc space for host copies of a, b, c and setup input values
    a = (int *)malloc(size); random_ints(a, N);
    b = (int *)malloc(size); random_ints(b, N);
    c = (int *)malloc(size);
```

Vector Addition on the Device: `main()`

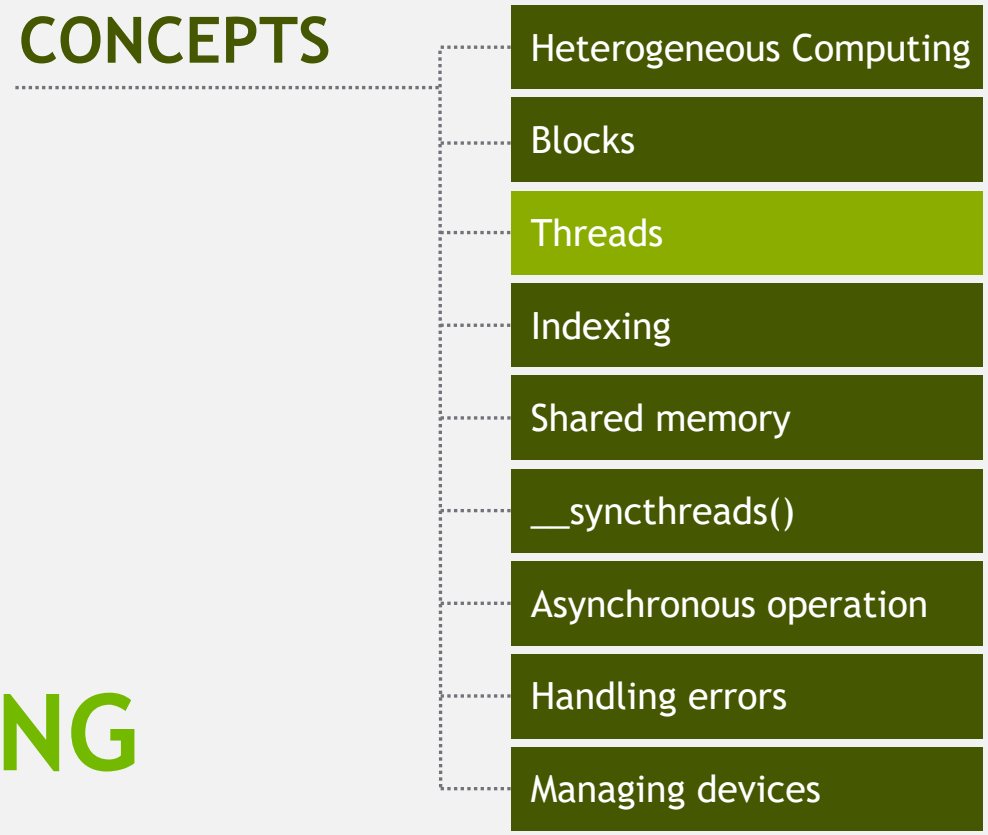
```
    // Copy inputs to device
    cudaMemcpy(d_a, a, size, cudaMemcpyHostToDevice);
    cudaMemcpy(d_b, b, size, cudaMemcpyHostToDevice);

    // Launch add() kernel on GPU with N blocks
    add<<<N,1>>>(d_a, d_b, d_c);

    // Copy result back to host
    cudaMemcpy(c, d_c, size, cudaMemcpyDeviceToHost);

    // Cleanup
    free(a);
    free(b);
    free(c);
    cudaFree(d_a);
    cudaFree(d_b);
    cudaFree(d_c);
    return 0;
}
```

INTRODUCING THREADS



Thread

CUDA Threads

- **Terminology:** a block can be split into parallel threads

Every thread has an ID.

- Let's change `add()` to use parallel *threads* instead of parallel *blocks*

```
__global__ void add(int *a, int *b, int *c) {
    c[threadIdx.x] = a[threadIdx.x] + b[threadIdx.x];
}
```

Vector Addition Using Threads: `main()`

```
#define N 512
int main(void) {
    int *a, *b, *c;                 // host copies of a, b, c
    int *d_a, *d_b, *d_c;           // device copies of a, b, c
    int size = N * sizeof(int);

    // Alloc space for device copies of a, b, c
    cudaMalloc((void **)&d_a, size);
    cudaMalloc((void **)&d_b, size);
    cudaMalloc((void **)&d_c, size);

    // Alloc space for host copies of a, b, c and setup input values
    a = (int *)malloc(size); random_ints(a, N);
    b = (int *)malloc(size); random_ints(b, N);
    c = (int *)malloc(size);
```

add<<<1,N>>>();

1 block
N threads of this block

Vector Addition Using Threads: `main()`

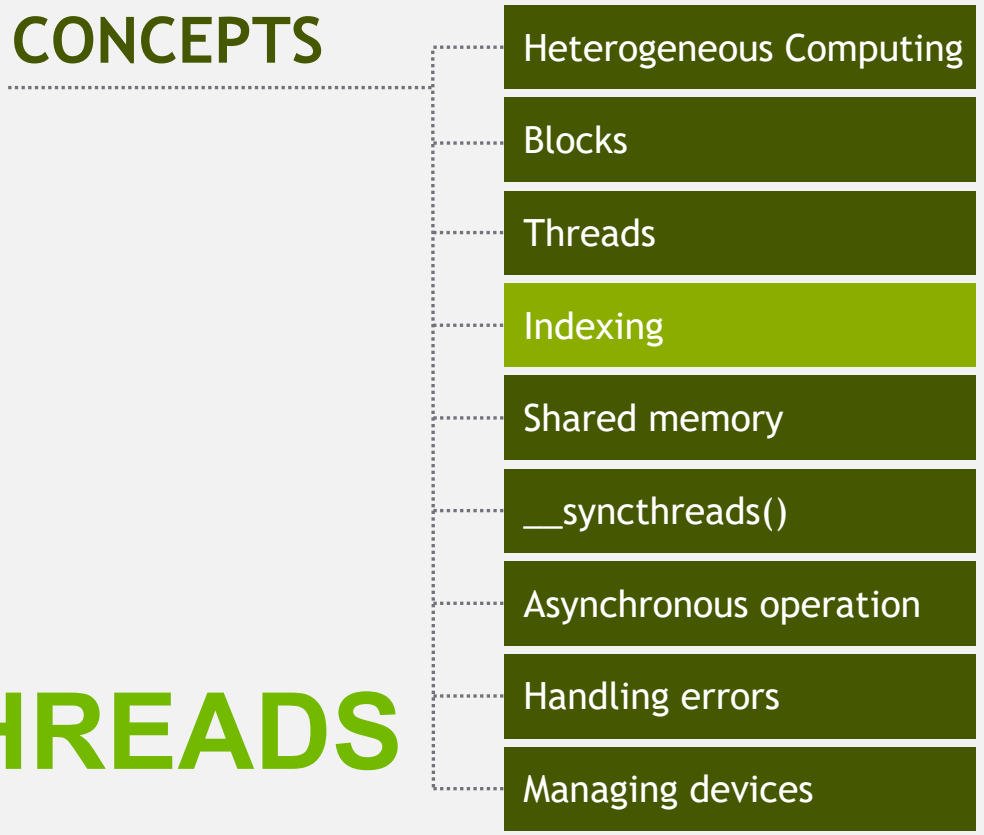
```
    // Copy inputs to device
    cudaMemcpy(d_a, a, size, cudaMemcpyHostToDevice);
    cudaMemcpy(d_b, b, size, cudaMemcpyHostToDevice);

    // Launch add() kernel on GPU with N threads
    add<<<1,N>>>(d_a, d_b, d_c);

    // Copy result back to host
    cudaMemcpy(c, d_c, size, cudaMemcpyDeviceToHost);

    // Cleanup
    free(a);
    free(b);
    free(c);
    cudaFree(d_a);
    cudaFree(d_b);
    cudaFree(d_c);
    return 0;
}
```

COMBINING THREADS AND BLOCKS



Indexing

Combining Blocks and Threads

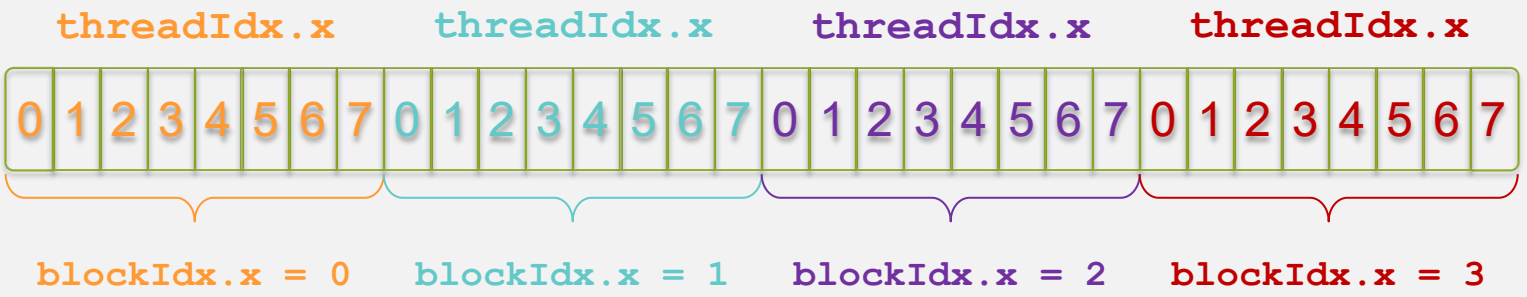
- We've seen parallel vector addition using:
  - Many blocks with one thread each
  - One block with many threads

- Let's adapt vector addition to use both blocks and threads

A GPU kernel is launched as a "Grid." The Grid is made of "Blocks," and each Block is made of individual "Threads."

Indexing Arrays with Blocks and Threads

- No longer as simple as using `blockIdx.x` and `threadIdx.x`
  - Consider indexing an array with one element per thread (8 threads/block)



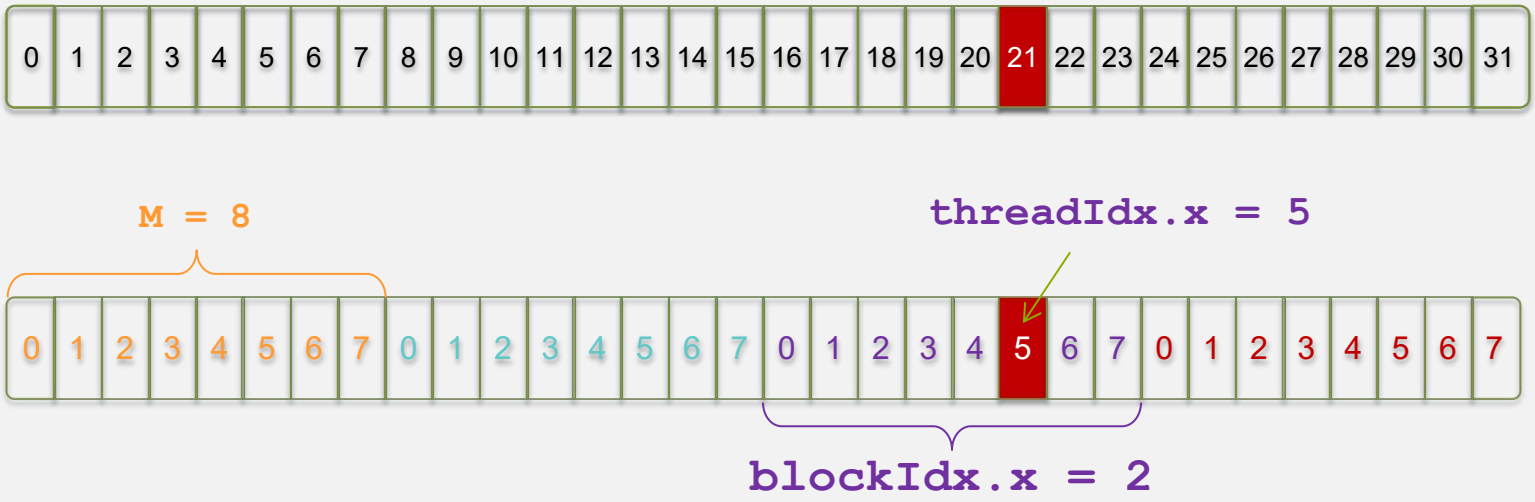
- With M threads/block a unique index for each thread is given by:

```
int index = threadIdx.x + blockIdx.x * M;
```

Every thread has a unique ID and belongs to a block

Indexing Arrays: Example

- Which thread will operate on the red element?




```
int index = threadIdx.x + blockIdx.x * M;
          =       5     +      2      * 8;
          = 21;
```

© NVIDIA 2013

Vector Addition with Blocks and Threads

- Use the built-in variable `blockDim.x` for threads per block

```
int index = threadIdx.x + blockIdx.x * blockDim.x;
```

- Combined version of `add()` to use parallel threads *and* parallel blocks

```
__global__ void add(int *a, int *b, int *c) {
    int index = threadIdx.x + blockIdx.x * blockDim.x;
    c[index] = a[index] + b[index];
}
```

Addition with Blocks and Threads: `main()`

```
#define N (2048*2048)
#define THREADS_PER_BLOCK 512
int main(void) {
    int *a, *b, *c;                 // host copies of a, b, c
    int *d_a, *d_b, *d_c;           // device copies of a, b, c
    int size = N * sizeof(int);

    // Alloc space for device copies of a, b, c
    cudaMalloc((void **)&d_a, size);
    cudaMalloc((void **)&d_b, size);
    cudaMalloc((void **)&d_c, size);

    // Alloc space for host copies of a, b, c and setup input values
    a = (int *)malloc(size); random_ints(a, N);
    b = (int *)malloc(size); random_ints(b, N);
    c = (int *)malloc(size);
```

© NVIDIA 2013

Addition with Blocks and Threads: `main()`

```
    // Copy inputs to device
    cudaMemcpy(d_a, a, size, cudaMemcpyHostToDevice);
    cudaMemcpy(d_b, b, size, cudaMemcpyHostToDevice);

    // Launch add() kernel on GPU
    add<<<N/THREADS_PER_BLOCK,THREADS_PER_BLOCK>>>(d_a, d_b, d_c);

    // Copy result back to host
    cudaMemcpy(c, d_c, size, cudaMemcpyDeviceToHost);

    // Cleanup
    free(a); free(b); free(c);
    cudaFree(d_a); cudaFree(d_b); cudaFree(d_c);
    return 0;
}
```

© NVIDIA 2013

Handling Arbitrary Vector Sizes

- Typical problems are not friendly multiples of `blockDim.x`

- Avoid accessing beyond the end of the arrays:

```
__global__ void add(int *a, int *b, int *c, int n) {
    int index = threadIdx.x + blockIdx.x * blockDim.x;
    if (index < n)
        c[index] = a[index] + b[index];
}
```

- Update the kernel launch:

```
add<<<(N + M-1) / M, M>>>(d_a, d_b, d_c, N);
```

Why Bother with Threads?

- Threads seem unnecessary
  - They add a level of complexity
  - What do we gain?

- Unlike parallel blocks, threads have mechanisms to:
  - Communicate — Threads in the same block can share information.
  - Synchronize — Threads can wait for each other.

© NVIDIA 2013